SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 30%

Duration : 90 Minutes

QUESTIONS: 5 **Total Marks:50**

Annexure A

Descriptive Written Test

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Descriptive Written Test

Instructions:

- Answer all questions.
- Each question carries 10 marks.
- Diagrams/tables should be used wherever applicable to support your answer.

1. Consider an autonomous vacuum-cleaning robot operating in a two-room environment, where each room can be either "clean" or "dirty," and the robot can move Left, Right, Suck, or do nothing.

a) Give the complete PEAS description (Performance measure, Environment, Actuators, Sensors) for this robot. *(5)*
b) Identify the type of intelligent agent (simple reflex, model-based, goal-based, or utility-based) best suited to control this robot, and justify your choice with reasoning specific to this scenario. *(5)*

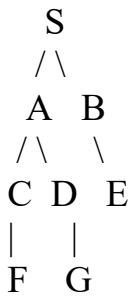
2. Analyze the following three task environments: (i) playing chess against a human opponent, (ii) an autonomous self-driving car navigating city traffic, and (iii) solving a crossword puzzle alone.

For each environment, identify and justify its properties with respect to: observability (fully/partially observable), determinism (deterministic/stochastic), episodic vs. sequential, static vs. dynamic, and discrete vs. continuous. Present your answer in a comparative table. *(10)*

3. A delivery robot must navigate from its starting location to a delivery point within a building represented as a grid, where certain cells are blocked by obstacles and movement is restricted to Up, Down, Left, and Right.

Formulate this scenario as a formal search problem by clearly defining: (i) Initial State, (ii) Goal State/Goal Test, (iii) Actions, (iv) Transition Model, and (v) Path Cost function. Justify each component with reference to the given scenario. *(10)*

4. Consider the following unweighted graph representing a road network, where S is the start node and G is the goal node:



Edges: S–A, S–B, A–C, A–D, B–E, C–F, E–G

Apply Breadth-First Search (BFS) to find a path from S to G. Show the frontier (queue) at each step of the search, and state the final path obtained along with its path cost.

Also explain the time complexity, space complexity, completeness, and optimality of BFS in general. *(10)*

5. Using the same graph given in Q4, apply Depth-First Search (DFS) to find a path from S to G. Show the stack at each step of the search, and state the final path obtained along with its path cost.

Compare the path found by DFS with the path found by BFS in Q4, and explain — with justification — which algorithm is more suitable for this problem and why. *(10)*

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand